Exponent Rules Maze ANSWER KEY

Start in the top-left room. Simplify or evaluate. Write answers with positive exponents only. Then go through the door with your answer.

START

1 Simplify.
$x^5 \cdot x^3$

x^{15}

Evaluate.
4^{-2}

x^{25}

Simplify.
$(x^2)^5$

$2x^{12}$

8 Simplify.
$(2x^3)^4$

x^8

-9

$5x^6$

-16

$\frac{1}{x^8}$

$\frac{1}{4x^4}$

$16x^{12}$

2 Evaluate.
2^{-3}

$-\frac{1}{8}$

Evaluate.
$3^2 \cdot 3^{-4}$

$\frac{1}{x^3}$

7 Simplify.
$\frac{x^2}{x^6}$

x^4

9 Simplify.
$\frac{12x^5y^2}{4x^7y}$

$\frac{1}{8}$

9

$\frac{1}{a^{10}}$

a^7

10

-8

$\frac{3y}{x^2}$

3 Simplify.
$(x^4)^3$

x^7

Simplify.
$a^5 \cdot a^{-2}$

13

6 Evaluate.
$(-3)^2 + 4^0$

9

10 Simplify.
$(3x^{-2}y^3)^2$

x^{12}

16

$5x^5y^2$

$6x^5y^2$

$5x^4$

-6

$\frac{9y^6}{x^4}$

4 Simplify.
$\frac{x^{10}}{x^5}$

x^5

5 Simplify.
$3x^2y \cdot 2x^3y$

$6x^6y^2$

Simplify.
$\frac{15x^8}{3x^2}$

$12x^6$

FINISH

You made it!

Exponent Rules Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **x^8**	$x^{5+3} \rightarrow x^8$
2	Answer: **$\frac{1}{8}$**	$2^{-3} = \frac{1}{2^3} \rightarrow \frac{1}{8}$
3	Answer: **x^{12}**	$x^{4 \cdot 3} \rightarrow x^{12}$
4	Answer: **x^5**	$x^{10-5} \rightarrow x^5$
5	Answer: **$6x^5y^2$**	$(3 \cdot 2)x^{2+3}y^{1+1} \rightarrow 6x^5y^2$
6	Answer: **10**	$9 + 1 \rightarrow 10$
7	Answer: **$\frac{1}{x^4}$**	$x^{2-6} = x^{-4} \rightarrow \frac{1}{x^4}$
8	Answer: **$16x^{12}$**	$2^4x^{3 \cdot 4} \rightarrow 16x^{12}$
9	Answer: **$\frac{3y}{x^2}$**	$\frac{12}{4}x^{5-7}y^{2-1} \rightarrow 3x^{-2}y \rightarrow \frac{3y}{x^2}$
10	Answer: **$\frac{9y^6}{x^4}$**	$3^2x^{-2 \cdot 2}y^{3 \cdot 2} \rightarrow 9x^{-4}y^6 \rightarrow \frac{9y^6}{x^4}$